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ABSTRACT

In 2021, Ö. Egecioğlu, V. Iršič introduced the concept of Fibonacci-run graph $\mathcal{R}_n$ as an induced subgraph of Hypercube. They conjectured that the diameter of $\mathcal{R}_n$ is given by $n - \lfloor (1 + \frac{n}{2})^{\frac{1}{2}} - \frac{3}{4} \rfloor$. In this paper, we introduce the concept of distance-barriers between vertices in $\mathcal{R}_n$ and provide a lower bound for the diameter of $\mathcal{R}_n$ via this concept. By constructing different types of distance-barriers, we show that the conjecture fails to hold for all $n \geq 230$ and for some n between 91 and 229. The lower bounds obtained turn out to be better than the result given in the conjecture.

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1. Introduction

Let s and t be integers such that $s < t$. For convenience, we use $[s : t]$ to denote the set $\{s, s + 1, \dots, t\}$ in this paper. For any real number a we use $\lfloor a \rfloor$ to denote the nearest integer less than or equal to a , and $\lceil a \rceil$ the nearest integer greater than or equal to a , respectively. Let $n \geq 1$ and

$$\mathcal{B}_n = \{b_1 b_2 \dots b_n \mid b_i \in \{0, 1\}, i \in [1 : n]\}.$$

Then any element of $\mathcal{B}_n$ is called a *binary string* (or simply a string) of length n . The *Hypercube* Q_n is the graph defined on the vertex set $\mathcal{B}_n$, two vertices $b_1 b_2 \dots b_n$ and $b'_1 b'_2 \dots b'_n$ being adjacent if $b_i \neq b'_i$ holds for exactly one $i \in [1 : n]$.

A well studied subfamily of Hypercubes called Fibonacci cubes were introduced by Hsu [6]. For $n \geq 2$, let

$$\mathcal{F}_n = \{b_1 b_2 \dots b_n \in \mathcal{B}_n \mid b_i b_{i+1} = 0, i \in [1 : n - 1]\}.$$

In other words, $\mathcal{F}_n$ is the set consisting of all strings of length n that have no two consecutive 1s. Then an element of $\mathcal{F}_n$ is called a *Fibonacci string*. The *Fibonacci cube* Γ_n has $\mathcal{F}_n$ as vertex set and two vertices are adjacent if they differ in exactly one coordinate. For convenience, we set $\Gamma_0 \cong K_1$ and $\Gamma_1 \cong K_2$. Fibonacci cubes have been studied from numerous points of view, as surveyed in [9]. Various families of subgraphs of Hypercubes have been introduced and studied, such as in [3,4,7,8,10–13,15–18].

Fibonacci-run graphs are a new family of graphs introduced by Egecioğlu and Iršič recently [1,2]. A string $\gamma \in \mathcal{B}_n$ is called a *run-constrained string* if every run of 1s in γ is immediately followed by a strictly longer run of 0s. Note that a run-constrained string of length $n \geq 2$ must end with 00. For $n \geq 0$, the *Fibonacci-run graph* $\mathcal{R}_n$ has the vertex set

$$V(\mathcal{R}_n) = \{r \mid r00 \text{ is a run-constrained binary string of length } n + 2\},$$

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and two vertices are adjacent if they differ in exactly one coordinate. Note that if $n = 0$, then $V(\mathcal{R}_n) = \{\lambda\}$ and so $\mathcal{R}_n \cong K_1$, where λ is the null string. If $n = 1$, then it is easily seen that $\mathcal{R}_1 \cong K_2$.

Although the Fibonacci-run graph $\mathcal{R}_n$ has the same number of vertices as the Fibonacci cube Γ_n , it has fewer edges than Γ_n , as well as some other different graph theoretical properties from Γ_n , as shown in papers [1,2]. As the study on $\mathcal{R}_n$ is in the initial stage, various crucial graph properties and features for $\mathcal{R}_n$ need to be determined. In [1,2], some interesting questions and conjectures on Fibonacci-run graphs have been proposed. For example, what is the number of vertices of degree k in $\mathcal{R}_n$ and what is the irregularity of $\mathcal{R}_n$? As for the hamiltonicity, it was conjectured that $\mathcal{R}_n$ is Hamiltonian if and only if $n \equiv 1 \pmod{3}$. As for the distance parameters of graph, they asked what is the radius Fibonacci-run graph $\mathcal{R}_n$ and conjectured that the diameter of $\mathcal{R}_n$ is given by:

Conjecture 1.1 ([1], Eq. (7)). Let $n \geq 0$. Then $\text{diam}(\mathcal{R}_n) = n - \lfloor \sqrt{1 + \frac{n}{2}} - \frac{3}{4} \rfloor$.

The problem of the radius of $\mathcal{R}_n$ has been completely solved in [14]. In the present paper, we focus on conjecture 1.1. We show that the above conjecture fails to hold for dimensions $n \in \{91, 94, 95, [119: 124], 131, 136, [152: 170], 172, 178, 181, 184, 185, [189: 223]\}$ and $n \geq 230$ in Section 4. The main tool used is the so called distance-barriers between vertices of $\mathcal{R}_n$, as introduced in Section 3.

In the rest of this section, we give some necessary notations and concepts. Let $\alpha \in \mathcal{B}_n$ be any string. Then we use $|\alpha|$ to denote the length of α . For convenience, we use the product to denote string concatenation, for example, 1^s is the string $11 \dots 1$ of length s . More generally, for a string α , α^s is the concatenation of s copies of α . We also take the null string λ into consideration. For examples, $1^0 = \lambda$ and in general, α^0 is also the null string. A non-extendable sequence of contiguous equal bits in a string is called a *run* (or *block*) of the string. For example, $1^3 0^2 1^2 0$ has 4 runs. A string f is called a *factor* of a string α if f appears as a sequence of $|f|$ consecutive bits of α .

2. Preliminaries

In this section, we introduce some distance-based concepts. Let G be a graph. Then the *distance* $d_G(u, v)$ (or simply $d(u, v)$) between vertices u and v of G is the length of a shortest path between them in G .

The *eccentricity* $e(v)$ of a vertex v in a connected graph G is the maximum distance from v and any other vertex, i.e.

$$e(v) = \max_{u \in V(G)} d(v, u).$$

The *radius* $\text{rad}(G)$ of G is the minimum eccentricity of the vertices of G , i.e. rad

$$\text{rad}(G) = \min_{v \in V(G)} e(v).$$

A vertex is called *central* if $e(v) = \text{rad}(G)$. The *center* $Z(G)$ of G is the set of all central vertices.

The radius and center of Fibonacci cube Γ_n and Fibonacci-run graph $\mathcal{R}_n$ were determined in [12,14], respectively. It is interesting to note that for a given $n \geq 0$, $\mathcal{R}_n$ and Γ_n have the same radius $\lceil \frac{n}{2} \rceil$.

The *diameter* $\text{diam}(G)$ of G is the maximal eccentricity $e(v)$ over all vertices v in G , i.e. diam

$$\text{diam}(G) = \max_{v \in V(G)} e(v) = \max_{u, v \in V(G)} d(u, v).$$

Hsu [6] proved that the diameter of Γ_n is n . Egecioglu and Iršič [1] conjectured the diameter of $\mathcal{R}_n$ is given by $n - \lfloor \sqrt{1 + \frac{n}{2}} - \frac{3}{4} \rfloor$ as stated in Conjecture 1.1.

The *Hamming distance* $H(\alpha, \beta)$ between $\alpha, \beta \in \mathcal{B}_n$ is the number of coordinates in which α and β differ. It is well known that $d_{Q_n}(\alpha, \beta) = H(\alpha, \beta)$ for any $\alpha, \beta \in V(Q_n)$ [5], and so for a subgraph G of Q_n , one always has $d_G(\alpha, \beta) \geq H(\alpha, \beta)$.

Let H and G be connected graphs. Then a mapping $f: V(H) \rightarrow V(G)$ is an *isometric embedding* if $d_H(u, v) = d_G(f(u), f(v))$ for any $u, v \in V(H)$. A *partial cube* is a connected graph which admits an isometric embedding into a Hypercube [5]. It is clear that the distance between any two vertices of a partial cube is exactly the Hamming distance between them.

It is shown that Γ_n is an isometric subgraph of Q_n [6], that is, Γ_n is a partial cube. However, Fibonacci-run graphs are in general not partial cubes as shown in Proposition 8.1 of [1] or Corollary 3.5 in the present paper, although there exist some particular vertices α, β of $\mathcal{R}_n$ with $d_{\mathcal{R}_n}(\alpha, \beta) = H(\alpha, \beta)$. In the following, we use an example to illustrate this.

Example 2.1. Let α and β be two vertices of $\mathcal{R}_{28}$:

$$\alpha = 1001111111000000001100011100 \text{ and}$$

$$\beta = 1110000100111000011111100000.$$

Then $H(\alpha, \beta) = 18$ and we can find a α, β -path of length 18 in $\mathcal{R}_n$ as follows:

$$\begin{aligned} &1001111111000000001100011100 \rightarrow 10001111111000000001100011100 \rightarrow \\ &1000011111000000001100011100 \rightarrow 10000011111000000001100011100 \rightarrow \\ &1000000111000000001100011100 \rightarrow 1000000110000000001100011100 \rightarrow \end{aligned}$$

1000000100000000001100011100 $\rightarrow$ 1000000100000000001100001100 $\rightarrow$
 1000000100000000001100001100 $\rightarrow$ 1000000100000000001100000000 $\rightarrow$
 1100000100000000001100000000 $\rightarrow$ 1110000100000000001100000000 $\rightarrow$
 1110000100100000001100000000 $\rightarrow$ 1110000100110000001100000000 $\rightarrow$
 1110000100111000001100000000 $\rightarrow$ 1110000100111000011100000000 $\rightarrow$
 1110000100111000011110000000 $\rightarrow$ 1110000100111000011111000000 $\rightarrow$
 1110000100111000011111100000.

From the above example we can see that such a shortest path is not a common *canonical* α, β -path (that is, a path where we first change each bit of α from 1 to 0 for which $a_i = 1$ and $b_i = 0$, and then change from 0 to 1 the other bits in which α and β differ) but a path obtained by using a particular order to change the bits in which they differ. However, a shortest path (lying completely in $\mathcal{R}_n$) whose length is the same as the Hamming distance does not always exist between vertices of $\mathcal{R}_n$ (as shown in Example 3.1). This will be discussed in the following section.

3. Distance-barrier of $\mathcal{R}_n$

This section contains two subsections. In the first subsection, we introduce the concept of distance-barrier between two vertices of $\mathcal{R}_n$, which will be used to give necessary and sufficient conditions for the existence of the shortest path whose length is the same as the Hamming distance between vertices of $\mathcal{R}_n$. In the second subsection, we apply those results to study $\text{diam}(\mathcal{R}_n)$.

3.1. Basic properties of distance-barriers

First we give an example to show that there is not always a shortest path whose length is the same as the Hamming distance between two vertices in $\mathcal{R}_n$.

Example 3.1. We choose the following two vertices α and β of $\mathcal{R}_{21}$:

$$\begin{aligned}\alpha &= a_1 a_2 \dots a_{20} a_{21} = 100110001110000011100 \text{ and} \\ \beta &= b_1 b_2 \dots b_{20} b_{21} = 111111111110000000000.\end{aligned}$$

It is clear that $H(\alpha, \beta) = 8$. Let $A = a_1 = 1$, $B = a_4 a_5 = 11$ and $C = a_9 a_{10} a_{11} = 111$ be the first three factors consisting of 1s of α . Then it can be seen that if we want to get a path between α and β lying completely in $\mathcal{R}_{21}$, we must change all the bits contained in at least two of A, B and C from 1 to 0 and then from 0 to 1. Since $|A| < |B| < |C|$, a shortest path between α and β (lying in $\mathcal{R}_{21}$) could be obtained by changing twice the bits in A and B as follows:

100110001110000011100 $\rightarrow$ 100110001110000001100 $\rightarrow$ 100110001110000000100 $\rightarrow$
 100110001110000000000 $\rightarrow$ 000110001110000000000 $\rightarrow$ 000010001110000000000 $\rightarrow$
 000000001110000000000 $\rightarrow$ 0000000011110000000000 $\rightarrow$ 000000111110000000000 $\rightarrow$
 000001111110000000000 $\rightarrow$ 000011111110000000000 $\rightarrow$ 000111111110000000000 $\rightarrow$
 001111111110000000000 $\rightarrow$ 011111111110000000000 $\rightarrow$ 111111111110000000000.

It is easily seen that the distance between α and β is

$$\begin{aligned}d_{\mathcal{R}_{21}}(\alpha, \beta) &= H(\alpha, \beta) + 2(|A| + |B|) \\ &= 8 + 2(1 + 2) \\ &= 14.\end{aligned}$$

From the above equation we know that $d_{\mathcal{R}_{21}}(\alpha, \beta) = 14$, which is not equal to $H(\alpha, \beta) = 8$.

For convenience, let $\alpha_{(k,l)}$ be the factor of string α beginning at the k th coordinate of α and of length l .

According to the characters of the factors $\alpha_{(1,11)} = 10011000111$ of α and $\beta_{(1,11)} = 11111111111$ of β in Example 3.1, we are motivated to define the following useful concept.

Definition 3.2. Let $\alpha = a_1 a_2 \dots a_n$ and $\beta = b_1 b_2 \dots b_n$ be two vertices of $\mathcal{R}_n$, and for some $k \geq 1, t \geq 1, s_i \geq 1$ for $i \in [1 : t]$ and $t_j \geq 1$ for $j \in [1 : t + 1]$,

$$\begin{aligned}\alpha_{(k,l)} &= 1^{r_1} 0^{s_1} 1^{r_2} \dots 0^{s_t} 1^{r_{t+1}} \text{ and} \\ \beta_{(k,l)} &= 1^{r_1} 1^{s_1} 1^{r_2} \dots 1^{s_t} 1^{r_{t+1}},\end{aligned}$$

where $l = \sum_{i=1}^t s_i + \sum_{j=1}^{t+1} r_j$. If there exists no $k' < k$ or $l' > l$ such that (1) and (2) hold: (1) $a_{k'} = 1, a_i = 0$ for $i \in [k' : k - 1] \setminus \{k'\}$, and $b_j = 1$ for $j \in [k' : k - 1]$, (2) $a_{k+l'-1} = 1, a_i = 0$ for $i \in [k + l : k + l' - 1] \setminus \{k + l' - 1\}$, and $b_j = 1$ for $j \in [k + l : k + l' - 1]$,

then we call $\begin{bmatrix} \alpha_{(k,l)} \\ \beta_{(k,l)} \end{bmatrix}$ a *distance-barrier of length l* between α and β with $t+1$ barriers beginning from the k th coordinate. In addition, we call 1^{r_i} the i th barrier with r_i being the *thickness* of barrier 1^{r_i} , $i \in [1 : t+1]$.

For example, by Definition 3.2, we know that $\begin{bmatrix} \alpha_{(1,11)} \\ \beta_{(1,11)} \end{bmatrix} = \begin{bmatrix} 10011000111 \\ 11111111111 \end{bmatrix}$ is a distance-barrier of length 11 between vertices α and β (in Example 3.1) of $\mathcal{R}_{21}$ with three barriers of thicknesses 1, 2 and 3, respectively.

Now we consider the effect of the distance-barriers between two vertices on the distance of them in the Fibonacci-run graph.

Let $\begin{bmatrix} \alpha_{(k,l)} \\ \beta_{(k,l)} \end{bmatrix} = \begin{bmatrix} 1^{r_1} 0^{s_1} 1^{r_2} \dots 0^{s_t} 1^{r_{t+1}} \\ 1^{r_1} 1^{s_1} 1^{r_2} \dots 1^{s_t} 1^{r_{t+1}} \end{bmatrix}$ be a distance-barrier between vertices α and β of $\mathcal{R}_n$, and $M_{(\alpha_{(k,l)}, \beta_{(k,l)})} = \max\{r_1, \dots, r_t, r_{t+1}\}$. Assume that $r_m = M_{(\alpha_{(k,l)}, \beta_{(k,l)})}$, $C_{(\alpha_{(k,l)}, \beta_{(k,l)})} = \sum_{i \in [1:t+1] \setminus m} r_i$ and $C'_{(\alpha_{(k,l)}, \beta_{(k,l)})} = \sum_{i \in [1:t+1]} r_i$. Then $C'_{(\alpha_{(k,l)}, \beta_{(k,l)})} - C_{(\alpha_{(k,l)}, \beta_{(k,l)})} = M_{(\alpha_{(k,l)}, \beta_{(k,l)})}$. Note that $C_{(\alpha_{(k,l)}, \beta_{(k,l)})} > 0$ since $t \geq 1$ and $r_i \geq 1$, $i \in [1 : t+1]$.

The following result is inspired from Example 3.1, and used to determine the length of a shortest path between two vertices having distance-barriers between them.

Lemma 3.3. Suppose that there are s (≥ 1) distance-barriers between vertices α and β of $\mathcal{R}_n$, and the i th distance-barrier is $\begin{bmatrix} \alpha_{(k_i, l_i)} \\ \beta_{(k_i, l_i)} \end{bmatrix}$, $i \in [1 : s]$. Then

$$\begin{aligned} d_{\mathcal{R}_n}(\alpha, \beta) &= H(\alpha, \beta) + 2 \sum_{i=1}^s C_{(\alpha_{(k_i, l_i)}, \beta_{(k_i, l_i)})} \\ &= H(\alpha, \beta) + 2 \sum_{i=1}^s (C'_{(\alpha_{(k_i, l_i)}, \beta_{(k_i, l_i)})} - M_{(\alpha_{(k_i, l_i)}, \beta_{(k_i, l_i)})}). \end{aligned}$$

Proof. First we consider the path between α and β obtained by only changing the coordinates of α and β in which they differ. Since there exists distance-barrier between α and β , there always is a string obtained by changing some bit (from 0 to 1) between two consecutive barriers that is not run-constrained. So such a path between α and β does not lie completely in $\mathcal{R}_n$. Now we must change some coordinates of α and β where they have the same bits in order to get a path between α and β lying completely in $\mathcal{R}_n$. It is obvious that such coordinates are some of those in the barriers, and they should be changed from 1 to 0 and then from 0 to 1. Suppose that a distance-barrier $\begin{bmatrix} \alpha_{(k_i, l_i)} \\ \beta_{(k_i, l_i)} \end{bmatrix}$ has $t_i + 1$ barriers, $i \in [1 : s]$. Then at least t_i barriers must be changed at least twice. Of course, we should choose the t_i barriers with the smallest thickness to get a shortest path between α and β . The desired result holds. $\square$

The following result is the main tool used in the present paper.

Lemma 3.4. Let α and β be two vertices of $\mathcal{R}_n$. Then $d_{\mathcal{R}_n}(\alpha, \beta) = H(\alpha, \beta)$ if and only if there is no distance-barrier between α and β .

Proof. If there is a distance-barrier between α and β , then $d_{\mathcal{R}_n}(\alpha, \beta) > H(\alpha, \beta)$ by Lemma 3.3. Suppose that there is no distance-barrier between $\alpha = a_1 a_2 \dots a_n$ and $\beta = b_1 b_2 \dots b_n \in V(\mathcal{R}_n)$, and $H(\alpha, \beta) = h$ and $a_{i_j} + b_{i_j} = 1$ for $j \in [1 : h]$. Now we show that, by induction on $H(\alpha, \beta)$, there is a α, β -path lying completely in $\mathcal{R}_n$ which can be obtained by only changing the coordinates in which α and β differ. Obviously, the length of such a path is $H(\alpha, \beta)$.

Obviously, the assertion holds for $H(\alpha, \beta) = 1$. Now we suppose that it holds for vertices with Hamming distance $h-1$ ($h \geq 2$ and so $i_1 < n$), that is, if there exists no distance-barrier between vertices with Hamming distance $h-1$, then there is a path with length $h-1$ lying in $\mathcal{R}_n$. Without loss of generality, suppose that $a_{i_1} = 0$ and so $b_{i_1} = 1$. Then there are the following three exclusive possibilities:

- (1) $i_1 = 1$,
- (2) $i_1 > 1$, $b_{i_1-1} = 0$ or $b_{i_1+1} = 0$, and
- (3) $i_1 > 1$, $b_{i_1-1} = 1$ and $b_{i_1+1} = 1$.

For the cases (1) and (2), let β' be the string obtained by changing the i_1 -th bit of β from 1 to 0. Then β' is run-constrained.

For the case (3), by the fact that there is not a distance-barrier between α and β , there exist $s \geq 1$ such that $a_{i_1+1} = \dots = a_{i_1+s} = 0$, $b_{i_1+1} = \dots = b_{i_1+s} = 1$, and $b_{i_1+s+1} = 0$. So the string β' obtained by changing the $(i_1 + s)$ -th bit of β from 1 to 0 is run-constrained.

In every of the above three cases, a run-constrained string β' is obtained. It is clear that $H(\alpha, \beta') = h-1$ and there is no distance-barrier between α and β' . Therefore, there is a shortest α, β' -path with length $h-1$ lying in $\mathcal{R}_n$. Hence we know that if there is no distance-barrier between α and β , then $d_{\mathcal{R}_n}(\alpha, \beta) = d_{\mathcal{R}_n}(\alpha, \beta') + d_{\mathcal{R}_n}(\beta', \beta) = H(\alpha, \beta') + 1 = H(\alpha, \beta)$. This completes the proof. $\square$

It is easily seen that the distance-barrier with the smallest length between vertices of $\mathcal{R}_n$ is $\begin{bmatrix} 1001 \\ 1111 \end{bmatrix}$. We immediately obtain an alternate proof of the following result.

Corollary 3.5 (2, Proposition 8.1). $\mathcal{R}_n$ is a partial cube if and only if $n \leq 6$.

Proof. Clearly, $n = 7$ is the smallest dimension such that there exists a distance-barrier between the vertices of $\mathcal{R}_n$. In fact, for $n \geq 7$, let $\alpha = 10010000^{n-7}$ and $\beta = 11110000^{n-7}$. Then both α and β are vertices of $\mathcal{R}_n$, and $\begin{bmatrix} \alpha_{(1,4)} \\ \beta_{(1,4)} \end{bmatrix} = \begin{bmatrix} 1001 \\ 1111 \end{bmatrix}$ is a distance-barrier between α and β . The desired result follows by Lemma 3.4. $\square$

3.2. Applications of distance-barriers

A lower bound for the diameter of $\mathcal{R}_n$ is given by Egecioğlu and Iršič ([1], Lemma 5.1 and Theorem 5.3). On the basis of this result and some calculation examples, they proposed Conjecture 1.1. In this section, firstly, a general lower bound for the diameter of $\mathcal{R}_n$ for all $n \in \mathbb{N}$ is obtained (Lemma 3.6). Then by constructing vertices with distance-barrier between them in various ways, we obtain some improved lower bounds for the diameter of $\mathcal{R}_n$ (see Lemmas 3.8–3.10).

Lemma 3.6. Let n be any non-negative integer. If $n \in [(2p^2 + 3p) : (2p^2 + 7p + 4)]$ for some $p \geq 0$, then $\text{diam}(\mathcal{R}_n) \geq n - p$.

Proof. Let $p \geq 0$ be an integer, $n = 2p^2 + 7p + 4$, and μ and ν be the following two vertices of $\mathcal{R}_n$:

$$\begin{aligned} \mu &= 10^2 1^3 \dots 1^{2p+1} 0^{2p+2} 1^{p+1} 0^p, \\ \nu &= 01^2 0^3 \dots 0^{2p+1} 1^{2p+2} 0^{p+1} 0^p. \end{aligned} \quad (3.1)$$

Note that if $p = 0$, then

$$\begin{aligned} \mu &= 1001, \\ \nu &= 0110; \end{aligned}$$

and if $p = 1$, then

$$\begin{aligned} \mu &= 1001110000110, \\ \nu &= 0110001111000. \end{aligned}$$

As there is no distance-barrier between μ and ν , $\text{diam}(\mathcal{R}_n) \geq d_{\mathcal{R}_n}(\mu, \nu) = H(\mu, \nu) = n - p$ by Lemma 3.4. Since any suffix of μ or ν is also a run-constrained string, we know that for any $n \leq 2p^2 + 7p + 4$, $\text{diam}(\mathcal{R}_n) \geq n - p$. By the fact that $2(p-1)^2 + 7(p-1) + 4 = 2p^2 + 3p - 1$, we know that the set of non-negative integers $\mathbb{N} = \bigsqcup_{p=0}^{+\infty} [(2p^2 + 3p) : (2p^2 + 7p + 4)]$. The result holds. $\square$

Note that the proof of Lemma 3.6 is same as the proof of Lemma 5.1 in [1], and it has shown that if $n = 2p^2 + 7p + 4$, then $\text{diam}(\mathcal{R}_n) \geq n - p$ there.

For convenience, the pair of vertices μ and ν (in Eq. (3.1)) or their suffixes of the same length n ($\leq 2p^2 + 7p + 4$) in the proof of Lemma 3.6 are called a pair of *H-type strings ending with p of length n* . The vertices μ and ν (of length $2p^2 + 7p + 4$) are called a pair of *complete H-type strings ending with p* . Otherwise, the *H-type strings* of length less than $2p^2 + 7p + 4$ are called *incomplete*. It is easily seen that the length of the longest run of the *H-type string* ending with p is $2(p+1)$. As there is no distance-barrier between a pair of *H-type strings*, the distance between a pair of *H-type strings* of length n in $\mathcal{R}_n$ is $n - p$ by Lemma 3.4.

The lower bound given in Lemma 3.6 is obtained by the distance between two vertices of $\mathcal{R}_n$ with no distance-barrier between them. By Lemma 3.3, we have every reason to believe that the bound could be improved. In the following, we first give an example to indicate that distance-barrier can be used as a tool in studying the diameter of $\mathcal{R}_n$. Then we give specific type of distance-barriers which will be used in the next section.

Example 3.7. Let $n = 91 = 2p^2 + 3p + 1$, where $p = 6$. Then by Lemma 3.6 we know for the following incomplete *H-type strings* ending with p

$$\begin{aligned} \mu &= 10^8 1^9 0^{10} 1^{11} 0^{12} 1^{13} 0^{14} 1^7 0^6 \text{ and} \\ \nu &= 01^8 0^9 1^{10} 0^{11} 1^{12} 0^{13} 1^{14} 0^7 0^6, \end{aligned}$$

$\text{diam}(\mathcal{R}_n) \geq d_{\mathcal{R}_n}(\mu, \nu) = n - 6 = 85$. If we chose vertices

$$\begin{aligned} \alpha &= (100)^5 11^{17} 0^{19} 1^{20} 0^{10} 0^9 \text{ and} \\ \beta &= (111)^5 10^{17} 1^{19} 0^{20} 1^{10} 0^9, \end{aligned}$$

then there is a distance-barrier $\begin{bmatrix} \alpha_{(1,16)} \\ \beta_{(1,16)} \end{bmatrix} = \begin{bmatrix} (100)^5 1 \\ (111)^5 1 \end{bmatrix}$ between α and β , and so $\text{diam}(\mathcal{R}_n) \geq d_{\mathcal{R}_n}(\mu, \nu) = H(\alpha, \beta) + 2C_{(\alpha_{(1,16)}, \beta_{(1,16)})} = 86$ by Lemma 3.3. The example ends here.

It can be seen that vertices α and β in Example 3.7 could be obtained from the following incomplete H -type strings, respectively:

$$0^{15}1^{18}0^{19}1^{20}0^{10}0^9 \text{ and} \\ 1^{15}0^{18}1^{19}0^{20}1^{10}0^9.$$

Inspired by this example, we give a type of distance-barriers between vertices in $\mathcal{R}_n$ which can be used to improve the lower bound for the diameter of $\mathcal{R}_n$, as given in the following result.

Lemma 3.8. Let $t \geq 2$, $s \in [0 : 2]$, $k \leq 3t + s$, $2 \leq b \leq t$ and

$$\mu = 0^k 1^{3t+s+1} 0^{3t+s+2} \dots 1^{3t+s+i-1} 0^{3t+s+i} \dots, \\ \nu = 1^k 0^{3t+s+1} 1^{3t+s+2} \dots 0^{3t+s+i-1} 1^{3t+s+i} \dots$$

be a pair of incomplete H -type strings, where $i \geq 2$ and the length of the longest run contained in μ (or ν) is at least $3t + s + \lfloor \frac{3b-k-4}{3(t-b)+s+1} \rfloor$. Then we can construct a pair of vertices such that there is a distance-barrier $\text{Bar}_1 = \begin{bmatrix} (100)^{b-1} 1 \\ (111)^{b-1} 1 \end{bmatrix}$ between them.

Proof. It can be seen that if $3b \leq k \leq 3t + s$, then for the following strings

$$\alpha = 0^{k-3b} (100)^b 1^{3t+s+1} 0^{3t+s+2} \dots \text{ and} \\ \beta = 1^{k-3b} (111)^b 0^{3t+s+1} 1^{3t+s+2} \dots,$$

Bar_1 is a distance-barrier between α and β .

If $k = 3b - 1$, then Bar_1 is a distance-barrier between strings

$$\alpha = 11000(100)^{b-2} 11^{3t+s-1} 0^{3t+s+2} \dots \text{ and} \\ \beta = 01111(111)^{b-2} 10^{3t+s-1} 1^{3t+s+2} \dots.$$

If $3b - 4 < k < 3b - 1$, then Bar_1 is a distance-barrier between strings

$$\alpha = 0^{k-3b+3} (100)^{b-1} 11^{3t+s} 0^{3t+s+2} \dots \text{ and} \\ \beta = 1^{k-3b+3} (111)^{b-1} 10^{3t+s} 1^{3t+s+2} \dots.$$

For the case that $1 \leq k \leq 3b - 4$, we first give a disjoint partition of $[1 : 3b - 4]$. Let $m = \lfloor \frac{3b-4}{3(t-b)+s+1} \rfloor$ and $I_i = [(3b - 4 - i - (i + 1)(3(t - b) + s)) : (3b - 4 - i - i(3(t - b) + s))]$, where $i = 0, 1, \dots, m$. In other words, starting from $3b - 4$, we divide $[1 : 3b - 4]$ into $m + 1$ subsets such that every subset consists of $3(t - b) + s$ consecutive numbers. Note that when $i = m$, $I_i = [1 : (3b - 4 - m - m(3(t - b) + s))]$. Then $[1 : 3b - 4] = \bigsqcup_{i=0}^m I_i$.

For $k \in I_0$, that is, $3b - 4 - [3(t - b) + s] \leq k \leq 3b - 4$, let

$$\alpha = (100)^{b-1} 11^{3(t-b)+k+s+3} 0^{3t+s+2} \dots, \\ \beta = (111)^{b-1} 10^{3(t-b)+k+s+3} 1^{3t+s+2} \dots.$$

Then Bar_1 is a distance-barrier between α and β .

If $k \in I_i$, $i \in [1 : m]$, without loss of generality, suppose that i is odd. Let

$$\alpha = (100)^{b-1} 11^{3b-1} 0^{3b+1} \dots 1^{3b+i-1} 0^{(i+1)(3(t-b)+s)+2i+k+4} \dots, \\ \beta = (111)^{b-1} 10^{3b-1} 1^{3b+1} \dots 0^{3b+i-1} 1^{(i+1)(3(t-b)+s)+2i+k+4} \dots.$$

Then Bar_1 is a distance-barrier between α and β . This completes the proof. $\square$

Lemma 3.8 implies that based on some pair of H -type strings of length long enough (note that it needs $\lfloor \frac{3b-k-4}{3(t-b)+s+1} \rfloor$ runs from the second run to the longest run), the strings having distance-barrier with b barriers can be constructed. For the case $s = 1$, $k = 3t$ and $b = t$, we can construct strings having distance-barrier with $b + 1$ barriers as shown in the following lemma.

Lemma 3.9. Let $t \geq 2$ and

$$\mu = 0^{3t} 1^{3t+3} 0^{3t+4} \dots, \text{ and}$$

$$v = 1^{3t} 0^{3t+3} 1^{3t+4} \dots$$

be a pair of incomplete H -type strings. Then a pair of vertices with distance-barrier $Bar_2 = \begin{bmatrix} (100)^t 1 \\ (111)^t 1 \end{bmatrix}$ between them can be constructed.

Proof. Let

$$\alpha = (100)^t 11^{3t+2} 0^{3t+4} \dots \text{ and}$$

$$\beta = (111)^t 10^{3t+2} 1^{3t+4} \dots$$

Then Bar_2 is a distance-barrier between α and β . $\square$

Suppose that for some $p \geq 0$, the length of the pair of incomplete H -type strings μ, v in Lemma 3.8 (or Lemma 3.9) is n , where $2p^2 + 3p \leq n \leq 2p^2 + 7p + 4$. Then we can assume that μ and v end with $p + m$ for some $m \geq 0$. So $d_{\mathcal{R}_n}(\mu, v) = H(\mu, v) = n - (p + m)$ by Lemma 3.4. For convenience, vertices α and β constructed on the basis of μ and v in the proof of Lemma 3.8 (or Lemma 3.9) are called NH -type strings ending with $p + m$. Let $\begin{bmatrix} \alpha_{(1,3r-2)} \\ \beta_{(1,3r-2)} \end{bmatrix}$ be the distance-barrier $\begin{bmatrix} (100)^{r-1} 1 \\ (111)^{r-1} 1 \end{bmatrix}$ between α and β ($r = b$ in Lemma 3.8 and $r = b + 1$ in Lemma 3.9). Then there are r barriers in $\begin{bmatrix} \alpha_{(1,3r-2)} \\ \beta_{(1,3r-2)} \end{bmatrix}$ and the thickness of every barrier is 1. It is easily seen that $H(\mu, v) = H(\alpha, \beta) + r$, and $C_{(\alpha_{(1,3r-2)}, \beta_{(1,3r-2)})} = r - 1$. So we could establish the relation between $d_{\mathcal{R}_n}(\alpha, \beta)$ and $d_{\mathcal{R}_n}(\mu, v)$ by Lemma 3.3:

$$\begin{aligned} d_{\mathcal{R}_n}(\alpha, \beta) &= H(\alpha, \beta) + 2C_{(\alpha_{(1,3r-2)}, \beta_{(1,3r-2)})} \\ &= H(\mu, v) - r + 2(r - 1) \\ &= d_{\mathcal{R}_n}(\mu, v) + r - 2 \\ &= n - (p + m) + r - 2 \\ &= (n - p) + r - (m + 2). \end{aligned} \tag{3.2}$$

More generally, we get the following result.

Lemma 3.10. Let μ and v be a pair of H -type strings, and α and β be a pair of NH -type strings obtained from μ and v . Suppose $\begin{bmatrix} \alpha_{(k_i, l_i)} \\ \beta_{(k_i, l_i)} \end{bmatrix}$ is the distance-barriers between vertices α and β , $i \in [1 : s]$. Then

$$\begin{aligned} d_{\mathcal{R}_n}(\alpha, \beta) &= d_{\mathcal{R}_n}(\mu, v) + \sum_{i=1}^s (C_{(\alpha_{(k_i, l_i)}, \beta_{(k_i, l_i)})} - M_{(\alpha_{(k_i, l_i)}, \beta_{(k_i, l_i)})}) \\ &= d_{\mathcal{R}_n}(\mu, v) + \sum_{i=1}^s (C'_{(\alpha_{(k_i, l_i)}, \beta_{(k_i, l_i)})} - 2M_{(\alpha_{(k_i, l_i)}, \beta_{(k_i, l_i)})}). \end{aligned}$$

Proof. As the barrier between α and β is obtained from μ and v by changing some bits of μ or v from 0 to 1, $H(\mu, v) = H(\alpha, \beta) + \sum_{i=1}^s C'_{(\alpha_{(k_i, l_i)}, \beta_{(k_i, l_i)})}$. By Lemma 3.4, $d_{\mathcal{R}_n}(\mu, v) = H(\mu, v)$ and $C'_{(\alpha_{(k_i, l_i)}, \beta_{(k_i, l_i)})} - C_{(\alpha_{(k_i, l_i)}, \beta_{(k_i, l_i)})} = M_{(\alpha_{(k_i, l_i)}, \beta_{(k_i, l_i)})}$. So

$$\begin{aligned} d_{\mathcal{R}_n}(\alpha, \beta) &= H(\alpha, \beta) + 2 \sum_{i=1}^s C_{(\alpha_{(k_i, l_i)}, \beta_{(k_i, l_i)})} \\ &= d_{\mathcal{R}_n}(\mu, v) - \sum_{i=1}^s C'_{(\alpha_{(k_i, l_i)}, \beta_{(k_i, l_i)})} + 2 \sum_{i=1}^s C_{(\alpha_{(k_i, l_i)}, \beta_{(k_i, l_i)})}, \end{aligned}$$

and the result follows. $\square$

Note that Eq. (3.2) is the case of Lemma 3.10 with only one distance-barrier ($s = 1$) having b barriers of thickness of 1 between α and β .

4. On the diameter of $\mathcal{R}_n$

In this section, we first show Conjecture 1.1 does not hold for almost all $n \in \mathbb{N}$. Then some exploring work on the diameter of $\mathcal{R}_n$ is carried out.

4.1. Conjecture 1.1 is in general not true

In this subsection, it is shown that when dimension $n \geq 91$, the result in Conjecture 1.1 comes to fail for some n , and for all $n \geq 230$, Conjecture 1.1 does not hold.

First, we show that diameter of $\mathcal{R}_n$ given in Conjecture 1.1 is nothing but the lower bound in Lemma 3.6, that is, the following result holds.

Lemma 4.1. Let $n \in [(2p^2 + 3p) : (2p^2 + 7p + 4)]$ for some $p \geq 0$. Then $\lfloor \sqrt{1 + \frac{n}{2}} - \frac{3}{4} \rfloor = p$.

Proof. Let $a > 0$ and $b \geq 0$. Then $\sqrt{a^2 + b} - a \geq 0$ and it is easily seen that the following result holds:

$$\sqrt{a^2 + b} - a \begin{cases} > 1, & b > 2a + 1, \\ = 1, & b = 2a + 1, \\ < 1, & b < 2a + 1. \end{cases}$$

Let $n \in [(2p^2 + 3p) : (2p^2 + 7p + 4)]$. Then $n = 2p^2 + 3p + k$ for some k such that $0 \leq k \leq 4p + 4$. Let $a = p + \frac{3}{4}$ and $b = \frac{k}{2} + \frac{7}{16}$. Then $b < p + 1 + \frac{7}{16} < 2p + \frac{3}{2} + 1 = 2a + 1$. As

$$\begin{aligned} \sqrt{1 + \frac{n}{2}} - \frac{3}{4} - p &= \sqrt{1 + \frac{2p^2 + 3p + k}{2}} - \left(\frac{3}{4} + p\right) \\ &= \sqrt{\left(p + \frac{3}{4}\right)^2 + \left(\frac{k}{2} + \frac{7}{16}\right)} - \left(\frac{3}{4} + p\right) \\ &= \sqrt{a^2 + b} - a, \end{aligned}$$

we know that $0 \leq \sqrt{1 + \frac{n}{2}} - \frac{3}{4} - p < 1$ by the above result, and so $\lfloor \sqrt{1 + \frac{n}{2}} - \frac{3}{4} \rfloor = p$. $\square$

By this result, for $n \in [(2p^2 + 3p) : (2p^2 + 7p + 4)]$, Conjecture 1.1 implies that $\text{diam}(\mathcal{R}_n) = n - p$, and so it means that the diameter of $\mathcal{R}_n$ has a liner rule with n in an interval of length $4p + 5 = (2p^2 + 7p + 4) - (2p^2 + 3p) + 1$ for every $p \geq 0$.

Let $p = 6$. Then $2p^2 + 3p + 1 = 91$. If Conjecture 1.1 is true, then $\text{diam}(\mathcal{R}_{91}) = n - p = 91 - 6 = 85$ by Lemma 4.1. However from Example 3.7 and Eq. (3.2), we know that for $n = 91$, $\text{diam}(\mathcal{R}_n) \geq n - (p + m) + r - 2 = 91 - (6 + 3) + 6 - 2 = 86$. This means that Conjecture 1.1 does not hold for $n = 91$. From Example 3.7, Eq. (3.2), and Lemma 4.1, we can see that if $n - p + r - (2 + m) > n - p$, that is $r > m + 2$, then $\text{diam}(\mathcal{R}_n) > n - p$, and so Conjecture 1.1 does not hold for $\mathcal{R}_n$. Based on this idea, we will show that for almost all $n \in \mathbb{N}$, Conjecture 1.1 does not hold. First, for a give p , we need to find the ranges of m and r such that $r > m + 2$.

Lemma 4.2. Let $p \geq 0$, $m \geq 0$ and n be an integer such that $2p^2 + 3p \leq n \leq 2p^2 + 7p + 4$. Suppose that μ and ν are H -type strings ending with $p + m$ of length n , and α and β be a pair of H -type strings obtained from μ and ν .

(1) If there is a distance-barrier Bar_1 between α and β for $b = m + 2 + e$ and $e \geq 1$, then $\lceil \frac{8p-37-\sqrt{64p^2-432p+89}}{10} \rceil \leq m \leq \lfloor \frac{8p-37+\sqrt{64p^2-432p+89}}{10} \rfloor$, and the maximum value of e is $\lfloor \frac{-6m-11+\sqrt{16m^2+(32p+56)m+32p+33}}{6} \rfloor$.

(2) If there is a distance-barrier Bar_2 between α and β for $b = m + 1 + e$ and $e \geq 1$, then $\lceil \frac{8p-31-\sqrt{64p^2-336p-79}}{10} \rceil \leq m \leq \lfloor \frac{8p-31+\sqrt{64p^2-336p-79}}{10} \rfloor$, and the maximum value of e is $\lfloor \frac{-6m-9+\sqrt{16m^2+(32p+56)m+32p+17}}{6} \rfloor$.

Proof. First we show that (1) holds. As there is a $\text{Bar}_1 = \left[\begin{smallmatrix} (100)^{b-1} 1 \\ (111)^{b-1} 1 \end{smallmatrix} \right]$ between α and β , the first run of the H -type strings μ (or ν) is a factor of 0^{3b} by Lemma 3.8. Let μ be such a string with the longest length. Then we can assume that

$$\begin{aligned} \mu &= 0^{3b} 1^{3b+1} 0^{3b+2} \dots, \text{ and} \\ \nu &= 1^{3b} 0^{3b+1} 1^{3b+2} \dots \end{aligned}$$

By the following two facts:

(i) the length of a complete H -type string ending with $p + m$ is $2(p + m)^2 + 7(p + m) + 4$ and the length of μ is n such that $2p^2 + 3p \leq n \leq 2p^2 + 7p + 4$;

(ii) μ is the suffix of a complete H -type string ending with $p + m$ which is obtained by deleting the factor $10^2 \dots 1^{3b-1}$ from this complete H -type string, we know that

$$(2(p + m)^2 + 7(p + m) + 4) - (2p^2 + 3p) \geq 1 + 2 + 3 + \dots + (3b - 1). \quad (a)$$

Let $b = m + 2 + e$ and $e \geq 1$. Then

$$9e^2 + (33 + 18m)e + (5m^2 + 19m - 8mp - 8p + 22) \leq 0 \quad (b)$$

by the above inequality (a). It is easily seen that the maximum e is the positive root such that the equality holds in inequality (b), and so $e \leq \frac{-6m-9+\sqrt{16m^2+(32p+56)m+32p+17}}{6}$, as desired.

To get the variation range of m , let $e = 1$. Then by the above inequality (b), we know that $5m^2 + (37 - 8p)m + (64 - 8p) \leq 0$, and so $\frac{8p-37-\sqrt{64p^2-432p+89}}{10} \leq m \leq \frac{8p-37+\sqrt{64p^2-432p+89}}{10}$, as desired.

Using the similar method, we can show that (2) holds. By Lemma 3.9, we know that

$$(2(p+m)^2 + 7(p+m) + 4) - (2p^2 + 3p) \geq 1 + 2 + 3 + \dots + 3b + (3b+1) + 2. \quad (c)$$

Let $b = m + 1 + e$, where $e \geq 1$. Then by inequality (c) we have

$$9e^2 + (27 + 18m)e + 5m^2 + (13 - 8p)m - 8p + 16 \leq 0. \quad (d)$$

From inequality (d), we know that $e \leq \frac{-6m-9+\sqrt{16m^2+(32p+56)m+32p+17}}{6}$, as desired.

Let $e = 1$. Then by inequality (d), we have $5m^2 + (31 - 8p)m + 52 - 8p \leq 0$. Therefore, $\frac{8p-31-\sqrt{64p^2-336p-79}}{10} \leq m \leq \frac{8p-31+\sqrt{64p^2-336p-79}}{10}$, as desired. $\square$

By Eq. (3.2), we know the value e in Lemma 4.2 is exactly the difference between $d_{\mathcal{R}_n}(\alpha, \beta)$ and $n - p$, that is, $e = d_{\mathcal{R}_n}(\alpha, \beta) - (n - p) = r - (m + 2)$, where r is the number of barriers contained in the distance-barrier between α and β .

Now we turn to consider Conjecture 1.1. For convenience, set

$$S_1 = \{n | n \geq 230\}, \text{ and}$$

$$S_2 = \{91, 94, 95, [119 : 124], 131, 136, [152 : 170], 172, 178, 181, 184, 185, [189 : 223]\}.$$

Theorem 4.3. Let $n \in S_1 \cup S_2$. Then $\text{diam}(\mathcal{R}_n) > n - \lfloor \sqrt{1 + \frac{n}{2}} - \frac{3}{4} \rfloor$.

Proof. By Lemma 4.1, $n - \lfloor \sqrt{1 + \frac{n}{2}} - \frac{3}{4} \rfloor = n - p$ if $2p^2 + 3p \leq n \leq 2p^2 + 7p + 4$. So we need to show that for $n \in S_1 \cup S_2$, $\text{diam}(\mathcal{R}_n) > n - p$. The basic idea of the proof is as follows. First, for a given p we find m such that there are H -type strings μ and ν ending with $p + m$ by Lemma 4.2. Then, based on μ and ν , the NH -type strings α and β can be constructed by Lemma 3.8 or 3.9. Thus, we know that $\text{diam}(\mathcal{R}_n) \geq d_{\mathcal{R}_n}(\alpha, \beta) > n - p$ by Eq. (3.2).

Now we show that the desired result holds for $n \in S_1$. It is clear that if $n \geq 2p^2 + 3p \geq 230$, then $p \geq 10$. For $p = 10$, we have $0 \leq m \leq 8$ by Lemma 4.2(1). Let $m = 2$ and $e = 1$. Then $b = m + 2 + e = 5$ and so we know that $n \in [(2(p+m)^2 + 7(p+m) + 4 - \sum_{k=1}^{3b} k - 1) : (2(p+m)^2 + 7(p+m) + 4 - \sum_{k=1}^{3b-1} k)]$ by inequality (a). By Lemma 3.8, the H -type strings can be chosen as ending with 12 ($= p + m$):

$$\begin{aligned} \mu_1 &= 0^{k_1} 1^{18} 0^{19} 1^{20} 0^{21} 1^{22} 0^{23} 1^{24} 0^{25} 1^{26} 0^{13} 0^{12} \text{ and} \\ \nu_1 &= 1^{k_1} 0^{18} 1^{19} 0^{20} 1^{21} 0^{22} 1^{23} 0^{24} 1^{25} 0^{26} 1^{13} 0^{12}, \\ \mu_2 &= 0^{k_2} 1^{17} 0^{18} 1^{19} 0^{20} 1^{21} 0^{22} 1^{23} 0^{24} 1^{25} 0^{26} 1^{13} 0^{12} \text{ and} \\ \nu_2 &= 1^{k_2} 0^{17} 1^{18} 0^{19} 1^{20} 0^{21} 1^{22} 0^{23} 1^{24} 0^{25} 1^{26} 0^{13} 0^{12}, \\ \mu_3 &= 0^{k_3} 1^{16} 0^{17} 1^{18} 0^{19} 1^{20} 0^{21} 1^{22} 0^{23} 1^{24} 0^{25} 1^{26} 0^{13} 0^{12} \text{ and} \\ \nu_3 &= 1^{k_3} 0^{16} 1^{17} 0^{18} 1^{19} 0^{20} 1^{21} 0^{22} 1^{23} 0^{24} 1^{25} 0^{26} 1^{13} 0^{12}, \end{aligned}$$

where $k_1 \in [7 : 17]$, $k_2 \in [1 : 16]$ and $k_3 \in [1 : 15]$ (so $230 \leq n \leq 240$, $241 \leq n \leq 256$ and $257 \leq n \leq 271$, respectively). By Lemma 3.8, for $i \in [1 : 3]$, NH -type strings α_i and β_i can be constructed from μ_i and ν_i such that there is a distance-barrier $\begin{bmatrix} (100)^{5-1} 1 \\ (111)^{5-1} 1 \end{bmatrix}$ between α_i and β_i . So for $230 \leq n \leq 271$, $\text{diam}(\mathcal{R}_n) \geq d_{\mathcal{R}_n}(\alpha, \beta) > n - p$ by Eq. (3.2).

Let $m = 3$ and $e = 1$. Then $b = m + 2 + e = 6$ and the H -type strings can be chosen as ending with 13 ($= p + m$):

$$\begin{aligned} \mu_4 &= 0^{k_4} 1^{19} 0^{20} 1^{21} 0^{22} 1^{23} 0^{24} 1^{25} 0^{26} 1^{27} 0^{28} 1^{14} 0^{13} \text{ and} \\ \nu_4 &= 1^{k_4} 0^{19} 1^{20} 0^{21} 1^{22} 0^{23} 1^{24} 0^{25} 1^{26} 0^{27} 1^{28} 0^{14} 0^{13}, \end{aligned}$$

where $k \in [10 : 12]$ (so $272 \leq n \leq 274$). By Lemma 3.8, NH -type strings α_4 and β_4 can be constructed from μ_4 and ν_4 such that there is a distance-barrier $\begin{bmatrix} (100)^{6-1} 1 \\ (111)^{6-1} 1 \end{bmatrix}$ between α_4 and β_4 . So for $272 \leq n \leq 274$, $\text{diam}(\mathcal{R}_n) \geq d_{\mathcal{R}_n}(\alpha, \beta) > n - p$ by Eq. (3.2).

For the case $p \geq 11$, we consider the suffix strings μ and ν of length n obtained from the complete H -type strings ending with $p + 2$, where $2p^2 + 7p + 4 \geq n \geq 2p^2 + 3p$. We will show that NH -type strings α and β can be constructed

from μ and ν such that there is a distance-barrier $\left[\begin{smallmatrix} (100)^{5-1}1 \\ (111)^{5-1}1 \end{smallmatrix} \right]$ between α and β . So we have $\text{diam}(\mathcal{R}_n) \geq d_{\mathcal{R}_n}(\alpha, \beta) = n - (p+2) + 5 - 2 = n - p + 1$ by Eq. (3.2). First, by the fact that $p \geq 11$ and $2(p+2)^2 + 7(p+2) + 4 - 2(p+1)^2 - 3(p+1) = 8p + 21 > \sum_{k=1}^{14} k = 105$, we know that if $2p^2 + 7p + 4 \geq n (\geq 2p^2 + 3p)$, then the length of the second run of μ more than 15. In other words, the first two runs of μ and ν are $0^k 1^{3t+s+1}$ and $1^k 0^{3t+s+1}$, respectively, where $k \geq 15$ and $s \in [0 : 2]$. By $\sum_{k=1}^{3t+s-1} k < 8p + 21$, we know that $t \leq \frac{1-2s+\sqrt{64p+169}}{6}$. Let $b = 5$. Then $\frac{3b-k-4}{3(t-b)+s+1} \leq \frac{15-1-4}{0+1} = 10$. This means that if we show that the length of the longest run of μ is at least $3t + s + 10$, then by Lemma 3.8, we can get a pair of NH-type strings α and β with the distance-barrier we desire between them. In fact, as μ and ν end with $p + 2$, the length of the longest run of μ is $2(p + 2 + 1) = 2p + 6$. Thus there is $2p + 6 - (3t + s + 1)$ runs behind the second run (of μ). Note that

$$\begin{aligned} 2p + 6 - (3t + s + 1) &\geq 2p + 6 - s - 1 - 3\left(\frac{1 - 2s + \sqrt{64p + 169}}{6}\right) \\ &= 2(p + 2) - \frac{1 + \sqrt{64p + 169}}{2}. \end{aligned}$$

Let $L(p) = 2(p + 2) - \frac{1 + \sqrt{64p + 169}}{2}$. Then $L(p)$ is increased with p and $L(11) \geq 11 > 10$. Thus we know that for $p \geq 11$ and $n \in [2p^2 + 3p : 2p^2 + 7p + 4]$, the pair of NH-type strings α and β of length n can be constructed from H-type strings μ and ν ending with $p + 2$, where there is a distance-barrier $\left[\begin{smallmatrix} (100)^{5-1}1 \\ (111)^{5-1}1 \end{smallmatrix} \right]$ between α and β . So $\text{diam}(\mathcal{R}_n) \geq d_{\mathcal{R}_n}(\alpha, \beta) = n - p + 1 > n - p$ for $n \geq 275 = 2 \times 11^2 + 3 \times 11$ by Eq. (3.2).

We can show that the bound also holds for $n \in S_2$. The proof is technical and is given in the Appendix. This completes the proof. $\square$

4.2. Further discussions on $\text{diam}(\mathcal{R}_n)$

In the above subsection, we show that Conjecture 1.1 is not true in general by improving the lower bound of the diameter of $\mathcal{R}_n$. To be more specific, for a graph $\mathcal{R}_n$, Conjecture 1.1 shows that $\text{diam}(\mathcal{R}_n) = n - p$ by Lemma 4.1. Theorem 4.3 shows that $\text{diam}(\mathcal{R}_n) \geq n - p + 1$ or $\text{diam}(\mathcal{R}_n) \geq n - p + 2$ for almost all integer $n \in \mathbb{N}$. The tool used is to find the distance-barriers with the thickness of every barrier being equal to 1. A more natural idea is to use the distance-barriers with barriers of thickness more than 1 to improve the lower bound of $\text{diam}(\mathcal{R}_n)$ and even to obtain $\text{diam}(\mathcal{R}_n)$. Some discussions on $\text{diam}(\mathcal{R}_n)$ are given along this line in this subsection.

In the proof of Theorem 4.3, we show that for $n \geq 275$, there is always distance-barrier $\left[\begin{smallmatrix} (100)^4 1 \\ (111)^4 1 \end{smallmatrix} \right]$ between some pairs of NH-type strings α and β and so we know that $d_{\mathcal{R}_n}(\alpha, \beta) = n - p + e$ for $e = 1$. By Lemmas 3.8 and 3.9, there may exist distance-barriers with more barriers between some pair of vertices α and β of $\mathcal{R}_n$ and so $d_{\mathcal{R}_n}(\alpha, \beta) = n - p + e$ for some $e > 1$.

By Lemma 4.2(1), for a given p we can find the variation range of $m (\geq 0)$ such that there is a distance-barrier $\left[\begin{smallmatrix} (100)^{m+2} 1 \\ (111)^{m+2} 1 \end{smallmatrix} \right]$ between the pair of NH-type strings α and β constructed by Lemma 3.8, and further for a given m , the maximum $e (\geq 1)$ can be obtained such that the distance-barrier $\left[\begin{smallmatrix} (100)^{m+1+e} 1 \\ (111)^{m+1+e} 1 \end{smallmatrix} \right]$ is contained in the pair of NH-type strings.

Let $i \in [1 : e]$. Then the maximum length of the NH-type strings α and β containing the distance-barrier $\left[\begin{smallmatrix} (100)^{m+1+i} 1 \\ (111)^{m+1+i} 1 \end{smallmatrix} \right]$ is $2(p + m)^2 + 7(p + m) + 4 - \sum_{k=1}^{3(m+2+i)-1} k$. Similarly, by Lemma 4.2 (2), for a given p we can calculate the m and e , and the length of the NH-type strings containing the distance-barrier $\left[\begin{smallmatrix} (100)^{m+i} 1 \\ (111)^{m+i} 1 \end{smallmatrix} \right]$ is $2(p + m)^2 + 7(p + m) + 4 - \sum_{k=1}^{3(m+1+i)-1} k - 2$, where $i \in [1 : e]$. Note that $e = d_{\mathcal{R}_n}(\alpha, \beta) - (n - p)$. So the higher this value e , the more lower bound of $\text{diam}(\mathcal{R}_n)$ can be improved.

In the following, first, for $p = 15$ we give an example to improve the lower bound of $\text{diam}(\mathcal{R}_n)$ by finding e with a higher value. Then by applying the same method, we get the result for $1 \leq p \leq 29$. To save space, we only list the result.

Example 4.4. Let $p = 15$ and $S = \{518, 523, 529, 532, 535, 536\}$. Then $\text{diam}(\mathcal{R}_n) \geq n - 13$ for $n \in [510 : 559] \setminus S$, and $\text{diam}(\mathcal{R}_n) \geq n - 12$ for $n \in [495 : 509] \cup S$.

Note that for $p = 15$, $\text{diam}(\mathcal{R}_n)$ should be $n - 15$ by Conjecture 1.1. The result shown in the above example can be obtained via the following analysis. By Lemma 4.2 (1), we calculate m , e and the corresponding maximum n as shown in Table 1. Note that we only consider $2p^2 + 3p (= 495) \leq n \leq 2p^2 + 7p + 4 (= 559)$, and let $n = 2p^2 + 7p + 4$ when $n > 2p^2 + 7p + 4$ in this process. By Table 1, we know that $e = 3$ for $495 \leq n \leq 509$ and $e = 2$ for $510 \leq n \leq 559$. By Lemma 4.2 (2), we calculate m , e and the corresponding n as shown in Table 2. Note that $495 \leq n \leq 559$, when $n > 559$ we set n is a non-number.

Table 1
 m and e when $p = 15$ by Lemma 4.2(1).

m	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
e	1	2	2	3	3	3	3	3	2	2	2	2	2	1	1	1	1	1
$n(\leq)$	523	523	548	502	508	509	505	496	559	559	559	548	523	559	559	559	547	506

Table 2
 m and e when $p = 15$ by Lemma 4.2(2).

m	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
e	2	2	3	3	3	3	3	3	3	3	2	2	2	2	2	1	1	1	1
$n(=)$	502	535	509	523	532	536	535	529	518	502	–	–	–	541	509	–	–	–	520

The contrast of Tables 1 and 2 shows that we only need to consider $e = 3$ for $n > 509$ in Table 2. So we know that $e = 3$ for $n = 518, 523, 529, 532, 535$ and 536 . This is the end of Example 4.4.

For $p \in [6 : 29]$, the results are shown in Table 3, where d denotes the diameter of $\mathcal{R}_n$. Note that for $p \in [6 : 29]$, $\text{diam}(\mathcal{R}_n) = n - p$ by Conjecture 1.1 if $2p^2 + 3p \leq n \leq 2p^2 + 7p + 4$. In Table 3, it is shown that for all $n \in [90 : 1889]$, the value of $e, e - 1$ or $e - 2$ is more than $n - p$.

Let $T = \bigcup_{i=1}^{21} T_i$. Then we have the following result by Table 3.

Corollary 4.5. Let $n \in [90 : 1889]$. Then

$$\text{diam}(\mathcal{R}_n) \geq \begin{cases} n + 1 - \lfloor \sqrt{\frac{81}{144} + \frac{5}{18}n} - \frac{13}{12} \rfloor, & n \in T; \\ n - \lfloor \sqrt{\frac{81}{144} + \frac{5}{18}n} - \frac{13}{12} \rfloor, & n \in [90 : 1889] \setminus T. \end{cases}$$

In the following example, some other types of distance-barriers are constructed to show that the lower bound of the diameter of $\mathcal{R}_n$ in Corollary 4.5 (or Table 3) could be further improved for some n .

Example 4.6. For the graph $\mathcal{R}_{288}$, $\text{diam}(\mathcal{R}_{288}) \geq 279 = 288 - 9$.

Note that $288 = 2 \times 11^2 + 7 \times 11 + 4 - 35$. By Conjecture 1.1, $\text{diam}(\mathcal{R}_{288}) = 288 - 11 = 277$. By Corollary 4.5 (or Table 3), we know that $\text{diam}(\mathcal{R}_{288}) \geq 288 - 10 = 278$. Now we show that $\text{diam}(\mathcal{R}_n) \geq 288 - 9 = 279$. Let

$$\begin{aligned} \mu &= 0^{35}1^{40}0^{41}1^{42}0^{43}1^{44}0^{22}0^{21} \text{ and} \\ \nu &= 1^{35}0^{40}1^{41}0^{42}1^{43}0^{44}1^{22}0^{21}. \end{aligned}$$

Obviously, μ and ν are H -type strings of length 288. Then we can construct the following NH -type strings from them:

$$\begin{aligned} \alpha &= (11000)^7 111^{38} 0^{41} 1^{42} 0^{43} 1^{44} 0^{22} 0^{21} \text{ and} \\ \beta &= (11111)^7 110^{38} 1^{41} 0^{42} 1^{43} 0^{44} 1^{22} 0^{21}. \end{aligned}$$

It is easily seen that there is a distance-barrier $\begin{bmatrix} \alpha_{(1,37)} \\ \beta_{(1,37)} \end{bmatrix} = \begin{bmatrix} (11000)^7 11 \\ (11111)^7 11 \end{bmatrix}$ between α and β . The thicknesses of all its barriers are 2. By Lemma 3.10, we have $d_{\mathcal{R}_{288}}(\alpha, \beta) = d_{\mathcal{R}_{288}}(\mu, \nu) + 2(8 - 2) = H(\mu\nu) + 12 = 288 - 21 + 12 = 288 - 9$. This is the end of Example 4.6.

Example 4.7. For the graph $\mathcal{R}_{520}$, $\text{diam}(\mathcal{R}_{520}) \geq 508 = 520 - 12$.

By Conjecture 1.1, we have $\text{diam}(\mathcal{R}_{520}) = d_{\mathcal{R}_{520}}(\mu, \nu) = 520 - 15 = 505$ since $520 = 2 \times 15^2 + 7 \times 15 + 4 - 39$. By Corollary 4.5 (or Table 3), we know that $\text{diam}(\mathcal{R}_{520}) \geq 520 - 13 = 507$. Let

$$\begin{aligned} \mu &= 0^{38}1^{39}0^{40}1^{41}0^{42}1^{43}0^{44}1^{45}0^{46}1^{47}0^{48}1^{24}0^{23} \text{ and} \\ \nu &= 1^{38}0^{39}1^{40}0^{41}1^{42}0^{43}1^{44}0^{45}1^{46}0^{47}1^{48}0^{24}0^{23}. \end{aligned}$$

Obviously, μ and ν are H -type strings of length 520. Then we can construct the following NH -type strings from them:

$$\begin{aligned} \alpha &= 100(11000)^7 1^{39} 0^{40} 1^{41} 0^{42} 1^{43} 0^{44} 1^{45} 0^{46} 1^{47} 0^{48} 1^{24} 0^{23} \text{ and} \\ \beta &= 111(11111)^7 0^{39} 1^{40} 0^{41} 1^{42} 0^{43} 1^{44} 0^{45} 1^{46} 0^{47} 1^{48} 0^{24} 0^{23}. \end{aligned}$$

It is easily seen that there is a distance-barrier $\begin{bmatrix} \alpha_{(1,35)} \\ \beta_{(1,35)} \end{bmatrix} = \begin{bmatrix} 100(11000)^6 11 \\ 111(11111)^6 11 \end{bmatrix}$ between α and β . The thicknesses of all its barriers are 2 except the first one is 1. By Lemma 3.10, we have $d_{\mathcal{R}_{520}}(\alpha, \beta) = d_{\mathcal{R}_{520}}(\mu, \nu) + (2(7 - 2) + 1) = H(\mu, \nu) + 12 = 520 - 23 + 11 = 520 - 12$.

Table 3A lower bound of the diameter of $\mathcal{R}_n$ when $n \in [90 : 1889]$.

$n \in d \geq$ p, e	$n - p + e$	$n - p + e - 1$	$n - p + e - 2$
6,1	$T_1 = \{91, 93, 94\}$	$[90:118] \setminus T_1$	
7,1	$T_2 = [119:124] \cup \{131, 136\}$	$[125:151] \setminus T_2$	
8,1	$T_3 = [152:170] \cup \{172, 178, 181, 184, 185\}$	$[171:188] \setminus T_3$	
9,1	$[189:223]$	$[224:229]$	
10,2	$T_4 = \{230, 232, 238, 239, 241\}$	$[230:274] \setminus T_4$	
11,2	$T_5 = [275 : 284] \cup \{230, 232, 238, 239, 241\}$	$[285:323] \setminus T_5$	
12,2	$T_6 = [324 : 352] \cup \{356, 361, 367, 370, 373, 374\}$	$[353:376] \setminus T_6$	
13,2	$[377:427]$	$[428:433]$	
14,3	$T_7 = \{436, 446, 451\}$	$[434:494] \setminus T_7$	
15,3	$T_8 = [495 : 509] \cup \{518, 523, 529, 532, 535, 536\}$	$[510:559] \setminus T_8$	
16,3	$T_9 = [560 : 598] \cup \{604, 607, 617, 619, 625, 626\}$	$[599:628] \setminus T_9$	
17,3	$[629:695]$	$[696:701]$	
18,4	$T_{10} = \{703, 706, 716, 718, 724, 725, 727\}$	$[702:778] \setminus T_{10}$	
19,4	$T_{11} = [779 : 799] \cup \{806, 815, 820, 826, 829, 832, 833\}$	$[800:859] \setminus T_{11}$	
20,4	$T_{12} = [860 : 910] \cup \{916, 931, 941, 945\}$	$[911:944] \setminus T_{12}$	
21,5	$T_{13} = \{946\}$	$[945:1028] \setminus T_{13}$	$[1029:1033]$
22,5	$T_{14} = \{1040, 1049, 1054, 1060, 1063, 1066, 1067\}$	$[1034:1126] \setminus T_{14}$	
23,5	$T_{15} = [1127:1153] \cup \{1157, 1171, 1174, 1192, 1193, 1195\}$	$[1154:1223] \setminus T_{15}$	
24,5	$T_{16} = [1224:1286] \cup \{1288, 1292, 1306, 1309, 1319, 1321\}$	$[1287:1324] \setminus T_{16}$	
25,6	$T_{17} = \{1327, 1328, 1330\}$	$[1325:1426] \setminus T_{17}$	$[1427:1429]$
26,6	$T_{18} = \{1438, 1445, 1454, 1459, 1465, 1468, 1471, 1472\}$	$[1430:1538] \setminus T_{18}$	
27,6	$T_{19} = [1539 : 1573] \cup \{1591, 1606, 1616, 1621\}$	$[1574:1651] \setminus T_{19}$	
28,6	$T_{20} = [1652 : 1727] \cup \{1732, 1744, 1751, 1760, 1765\}$	$[1728:1768] \setminus T_{20}$	
29,7	$T_{21} = \{1771, 1774, 1777, 1778\}$	$[1769:1888] \setminus T_{21}$	1889

Those examples show that we need to consider all the possible distance-barriers between vertices of $\mathcal{R}_n$ to find the diameter of $\mathcal{R}_n$. In fact, we can find a lower bound of $\text{diam}(\mathcal{R}_n)$ for a given distance-barrier using the method as given in Lemma 4.2. However, there are many types of distance-barriers which are effective to the (maximum) distance of $\mathcal{R}_n$, as shown in Examples 4.6 and 4.7. We believe it is not an easy task to find such distance-barriers.

Our extensive study seems to indicate that there is no unified formula for $\text{diam}(\mathcal{R}_n)$, and the following problem still remains open.

Question 4.8. What is the diameter of $\mathcal{R}_n$?

Appendix

The proof of Theorem 4.3 for $n \in S_2$ is given in this appendix.

First we consider the case $n \in \{91, 94, 95, 131, 136, 172, 178, 181, 184, 185\}$. For a given p , we can give H -type strings μ and ν ending with $p + m$ by Lemma 4.2 (2). By Lemma 3.9, NH -type strings α and β can be constructed from μ and ν , and for $b = m + 2$, there is a distance-barrier $\begin{bmatrix} (100)^b 1 \\ (111)^b 1 \end{bmatrix}$ between α and β . So from (c), $\text{diam}(\mathcal{R}_n) > n - p$ holds for $n = 2(p + m)^2 + 7(p + m) + 4 - \sum_{k=1}^{b+1} k - 2$. For $p \leq 5$, we find there is no such m . For $p = 6$, we have $m = 1, 2$ and 3 . Therefore $n = 94, 95$ and 91 . For $n = 94$, the H -type strings (ending with $p + m = 6 + 1 = 7$) are

$$\begin{aligned} \mu &= 0^9 1^{12} 0^{13} 1^{14} 0^{15} 1^{16} 0^8 0^7 \text{ and} \\ \nu &= 1^9 0^{12} 1^{13} 0^{14} 1^{15} 0^{16} 1^8 0^7. \end{aligned}$$

By Lemma 3.9, the NH -type strings based on μ and ν can be constructed as follows

$$\begin{aligned} \alpha &= (100)^3 10^{11} 0^{13} 1^{14} 0^{15} 1^{16} 0^8 0^7 \text{ and} \\ \beta &= (111)^3 10^{11} 1^{13} 0^{14} 1^{15} 0^{16} 1^8 0^7. \end{aligned}$$

Note that there are $r = 4$ barriers contained in the distance-barrier between α and β and so $e = r - (m + 2) = 1$. So by Eq. (3.2), $\text{diam}(\mathcal{R}_n) \geq d_{\mathcal{R}_n}(\alpha, \beta) = (n - p) + e > n - p = 88$. Similarly, for $n = 95, 91$ it can be shown that $e = 1$, and so $\text{diam}(\mathcal{R}_n) \geq d_{\mathcal{R}_n}(\alpha, \beta) = (n - p) + e > n - p$. Note that the case $n = 91$ is shown in Example 3.7. To save space, for other p , we only give the m and n such that $e \geq 1$, but omit listing the H -type strings and the corresponding NH -type strings. For $p = 7$, by Lemma 4.2 (2) we get $m \in [0 : 5]$. If we choose $m = 1$ and 2 , then $n = 131$ and 136 , respectively. (Note that if $m = 0, 3, 4$ or 5 , then $n = 121, 136, 131$ or 121 , respectively. Obviously, everyone of those numbers of n is contained in some interval of $S_1 \cup S_2$, or the same one as the corresponding value of some m that has been list. The same applies to the other cases.) For $p = 8, m \in [0 : 6]$ by Lemma 4.2 (2). Let $m = 1, 2, 3, 4$ and 5 . Then $n = 172, 178, 181, 184$ and 185 .

Next, we show that the result holds for other integers in S_2 . For a given p , we will first find the m such that $n - p + b - (m + 2) > n - p$ by Lemma 4.2 (1), where $b = m + 3$. Then we know that $n \in [(2(p+m)^2 + 7(p+m) + 4 - \sum_{k=1}^{3b} k - 1) : (2(p+m)^2 + 7(p+m) + 4 - \sum_{k=1}^{3b-1} k)]$ by inequality (a), and so by Lemma 3.8, the pair of NH -type strings can be constructed. For $p \leq 6$, we find there is not m such that $n - (p + m) + b - 2 > n - p$. For $p = 7$, we have $m \in [1 : 3]$. If $m = 2$, then $b = m + 3 = 5$. For $k \in [10 : 15]$ (and so $119 \leq n \leq 124$) we chose H -type strings ending with $p + m = 9$:

$$\begin{aligned}\mu &= 0^k 1^{16} 0^{17} 1^{18} 0^{19} 1^{20} 0^{10} 0^9, \text{ and} \\ \nu &= 1^k 0^{16} 1^{17} 0^{18} 1^{19} 0^{20} 1^{10} 0^9.\end{aligned}$$

Then by Lemma 3.8, NH -type strings α_i and β_i containing distance-barrier Bar_1 can be constructed as follows for $k = i + 9$, $i \in [1 : 6]$:

$$\begin{aligned}\alpha_1 &= (100)^4 1^{14} 0^{16} 1^{18} 0^{19} 1^{20} 0^{10} 0^9 \text{ and } \beta_1 = (111)^4 10^{14} 1^{16} 0^{18} 1^{19} 0^{20} 1^{10} 0^9, \\ \alpha_2 &= (100)^4 1^{14} 0^{17} 1^{18} 0^{19} 1^{20} 0^{10} 0^9 \text{ and } \beta_2 = (111)^4 10^{14} 1^{17} 0^{18} 1^{19} 0^{20} 1^{10} 0^9, \\ \alpha_3 &= (100)^4 1^{15} 0^{17} 1^{18} 0^{19} 1^{20} 0^{10} 0^9 \text{ and } \beta_3 = (111)^4 10^{15} 1^{17} 0^{18} 1^{19} 0^{20} 1^{10} 0^9, \\ \alpha_4 &= 0(100)^4 1^{15} 0^{17} 1^{18} 0^{19} 1^{20} 0^{10} 0^9 \text{ and } \beta_4 = 1(111)^4 10^{15} 1^{17} 0^{18} 1^{19} 0^{20} 1^{10} 0^9, \\ \alpha_5 &= 11000(100)^3 1^{15} 0^{17} 1^{18} 0^{19} 1^{20} 0^{10} 0^9 \text{ and } \beta_5 = 01111(111)^3 10^{15} 1^{17} 0^{18} 1^{19} 0^{20} 1^{10} 0^9, \\ \alpha_6 &= (100)^5 1^{16} 0^{17} 1^{18} 0^{19} 1^{20} 0^{10} 0^9 \text{ and } \beta_6 = (111)^5 0^{16} 1^{17} 0^{18} 1^{19} 0^{20} 1^{10} 0^9.\end{aligned}$$

So we have $b = 5$ and so $e = b - (m + 2) = 1$. Therefore, $d_{\mathcal{R}_n}(\alpha, \beta) = n - p + e = n - p + 1 > n - p$ for $n \in [119 : 124]$.

Similarly, we can consider the cases that $p = 8$ and 9 . To save space, we only list the H -type strings which can be obtained by Lemma 4.2(1), and then the NH -type strings which contain distance-barrier Bar_1 can be constructed by Lemma 3.8.

For $p = 8$, we have $m \in [1 : 3]$. Let $m = 1$ and so $b = m + 3 = 4$. We can chose H -type strings ending with $9 (= p + m)$:

$$\begin{aligned}\mu &= 0^k 1^{13} 0^{14} 1^{15} 0^{16} 1^{17} 0^{18} 1^{19} 0^{20} 1^{10} 0^9 \text{ and} \\ \nu &= 1^k 0^{13} 1^{14} 0^{15} 1^{16} 0^{17} 1^{18} 0^{19} 1^{20} 0^{10} 0^9,\end{aligned}$$

where $k \in [1 : 12]$ (and so $152 \leq n \leq 163$). Let $m = 3$. Then $b = m + 3 = 6$ and we can chose H -type strings ending with $11 (= p + m)$:

$$\begin{aligned}\mu &= 0^k 1^{19} 0^{20} 1^{21} 0^{22} 1^{23} 0^{24} 1^{12} 0^{11} \text{ and} \\ \nu &= 1^k 0^{19} 1^{20} 0^{21} 1^{22} 0^{23} 1^{24} 0^{12} 0^{11},\end{aligned}$$

where $k \in [12 : 18]$ (and so $164 \leq n \leq 170$). So we have $\text{diam}(\mathcal{R}_n) \geq n - 7 > n - 8$ for $n \in [152 : 170]$.

For $p = 9$, we have $m \in [0 : 7]$. Let $m = 2$. Then $b = m + 3 = 5$ and the H -type strings can be chosen as ending with $11 (= p + m)$:

$$\begin{aligned}\mu_1 &= 0^{k_1} 1^{17} 0^{18} 1^{19} 0^{20} 1^{21} 0^{22} 1^{23} 0^{24} 1^{12} 0^{11} \text{ and} \\ \nu_1 &= 1^{k_1} 0^{17} 1^{18} 0^{19} 1^{20} 0^{21} 1^{22} 0^{23} 1^{24} 0^{12} 0^{11}, \\ \mu_2 &= 0^{k_2} 1^{16} 0^{17} 1^{18} 0^{19} 1^{20} 0^{21} 1^{22} 0^{23} 1^{24} 0^{12} 0^{11} \text{ and} \\ \nu_2 &= 1^{k_2} 0^{16} 1^{17} 0^{18} 1^{19} 0^{20} 1^{21} 0^{22} 1^{23} 0^{24} 1^{12} 0^{11},\end{aligned}$$

where $k_1 \in [2 : 16]$ and $k_2 \in [1 : 15]$ (so $189 \leq n \leq 203$ and $204 \leq n \leq 218$). Let $m = 3$. Then $b = m + 3 = 6$ and the H -type strings can be chosen as ending with $12 (= p + m)$:

$$\begin{aligned}\mu &= 0^k 1^{19} 0^{20} 1^{21} 0^{22} 1^{23} 0^{24} 1^{25} 0^{26} 1^{13} 0^{12} \text{ and} \\ \nu &= 1^k 0^{19} 1^{20} 0^{21} 1^{22} 0^{23} 1^{24} 0^{25} 1^{26} 0^{13} 0^{12},\end{aligned}$$

where $k \in [14 : 18]$ (so $219 \leq n \leq 223$). $\square$

Data availability

No data was used for the research described in the article.

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